
EXPLORE THE ATLAS

Why and how to leverage
the actor statement corpus

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Summary

1. What is a semantic map?
2. Where does it come from? ...*a text embedding model*
3. The corpus of actor statements is post-demographic
4. What to do: just observe
5. How the map supports exploration
6. Issues vs. clusters
7. Notebooks demo

What is a semantic map?

Each dot is a statement.

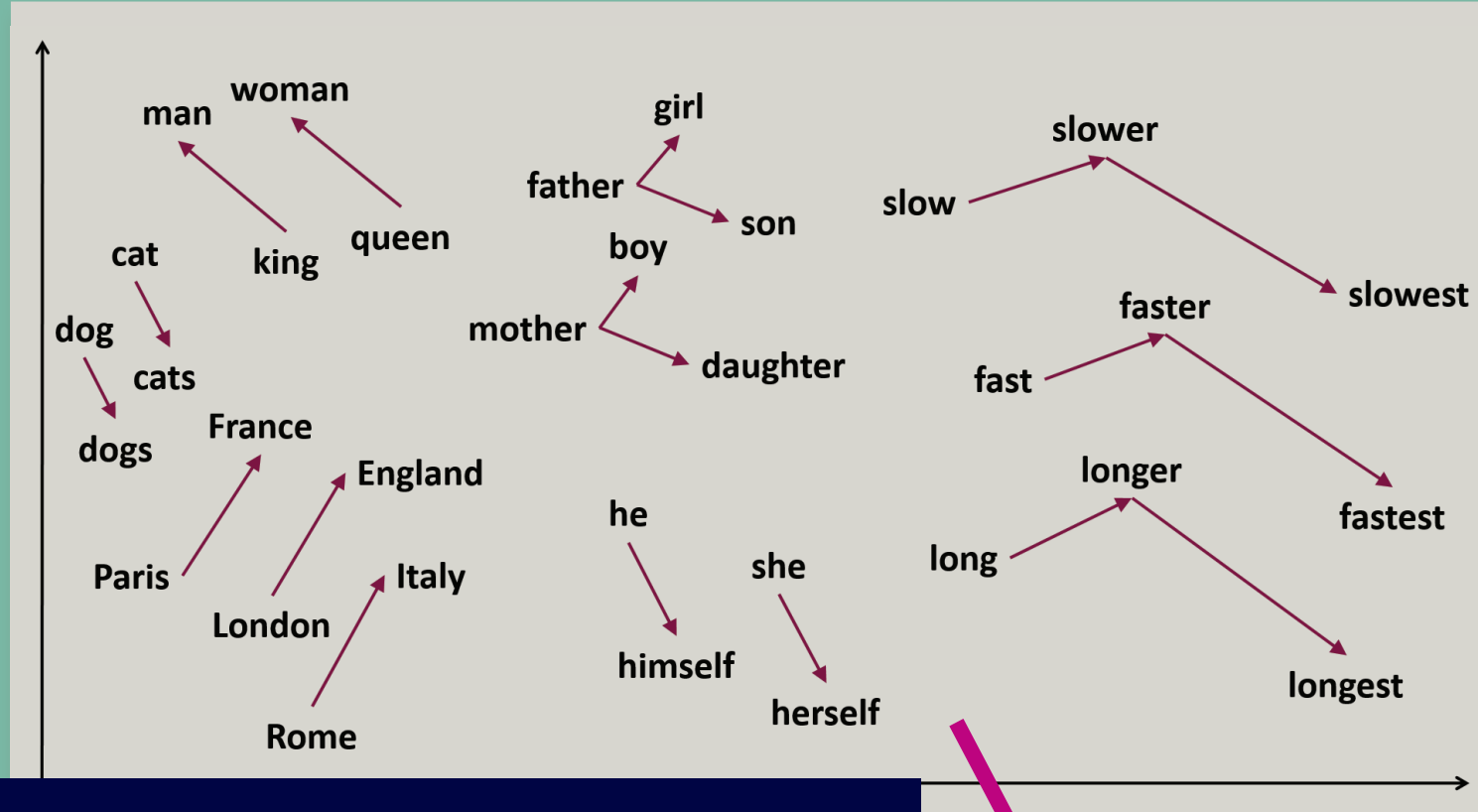
Close = similar
Distant = dissimilar
(on a semantic level)

Clusters are groups
of very similar
documents.

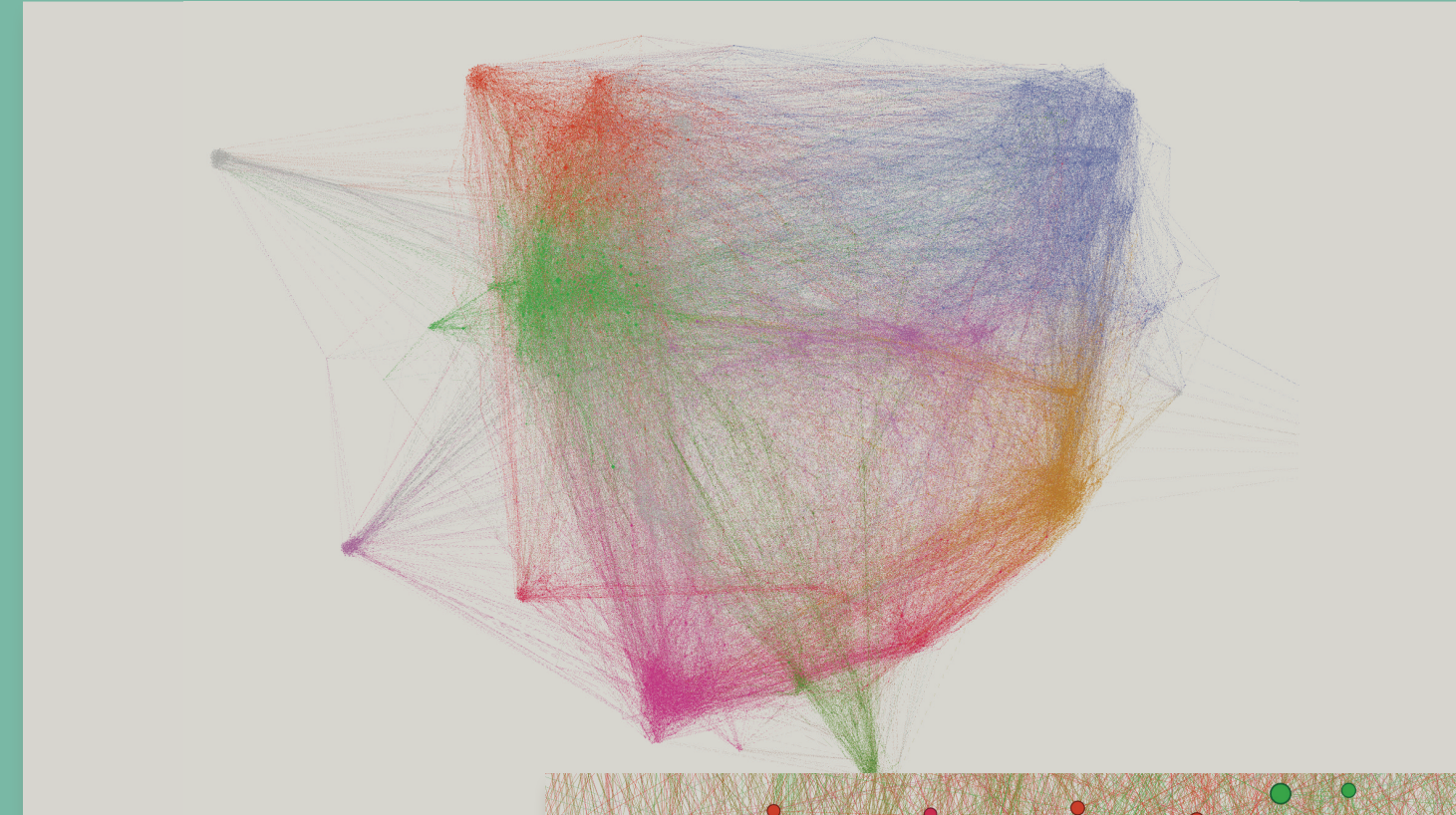


The map is then
rendered to facilitate
the identification
of clusters, and
those are qualified
manually.

Where does the map come from? *A text embedding model.*



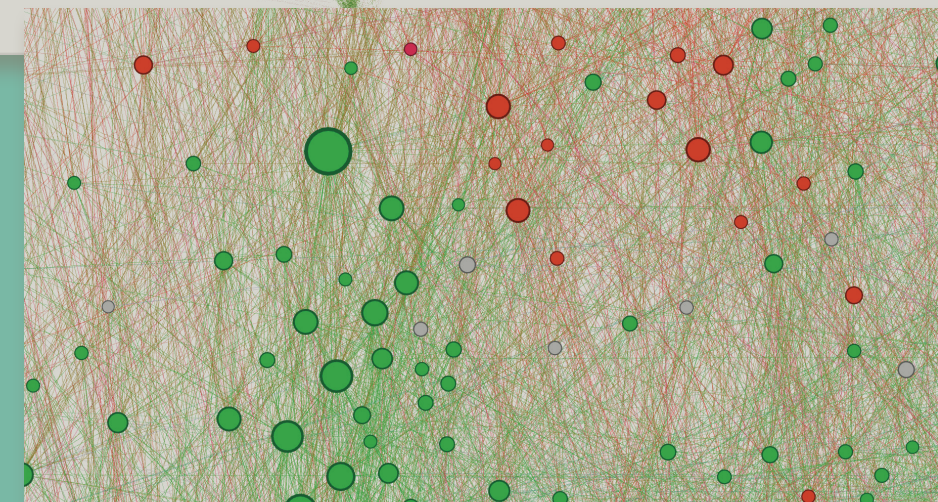
Word2Vec is a word embedding model: it transforms words into vectors preserving relations of meaning.



The vectors are flattened into a map through “dimensionality reduction”.



<https://pair-code.github.io/understanding-umap/>



We use a text embedding that is similar, but with entire text documents.

Post-demographics

Is online friendship “real” friendship?

Are social media accounts “real” people?

How to study data from the internet if it is not representative?



Richard Rogers

“Conceptually, with the ‘post’ prefixed to demographics, the idea is to stand in contrast to how the study of demographics organizes groups, markets and voters in a sociological sense.”

—Rogers, R. (2009). *Post-demographic machines*. *Walled garden*, 38(2009), 29-39.

Takeaways:

We do not have to pretend that a user account matches a person, which is not true in general.

We study actor statements for what they are, regardless of their lack of “representativity”.

What to do with the dataset? “*Just observe.*”

Don't take sides.

“Just observe
and describe
controversies”

- Do not restrain your observation to any single theory or methodology;
- observe from as many viewpoints as possible;
- listen to actors' voices more than to your own presumptions.

—Venturini, T. (2010). *Diving in magma: how to explore controversies with actor-network theory*. *Public understanding of science*, 19(3), 258-273.



Bruno Latour

What makes it hard
to “just observe” is
precisely this “just”.

Don't give your
opinion or tell who is
right or wrong.

What to do with the dataset? “*Just observe.*”

“Just observe
and describe
controversies”



Bruno Latour

Which statements “just observe”?

1. Climate change is real.
2. Climate change is not real.
3. According to Donald J. Trump, climate change is not real.
4. Many major oil companies now publicly acknowledge human-caused climate change while still advocating for continued fossil fuel use during an “energy transition,” according to the United States Senate Committee.

Takeaway:

Assess your own statements
(for the policy brief) and
check you “just” observe.

What to do with the dataset? “*Just observe.*”

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Bruno Latour

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Second-degree objectivity

If there is no objectivity in controversies, how can they be studied objectively? Would it not be wiser to wait?

How can we be objective when no one else is?

“Latour calls ‘second-degree objectivity’ the effort to consider as much subjectivity as possible. Unlike first-degree objectivity, which defines a situation of collective agreement, second-degree objectivity is attained by **revealing the full extent of actors’ disagreement and is thereby typical of controversial settings.”**

—Venturini, T. (2010). *Diving in magma: how to explore controversies with actor-network theory*. *Public understanding of science*, 19(3), 258-273.

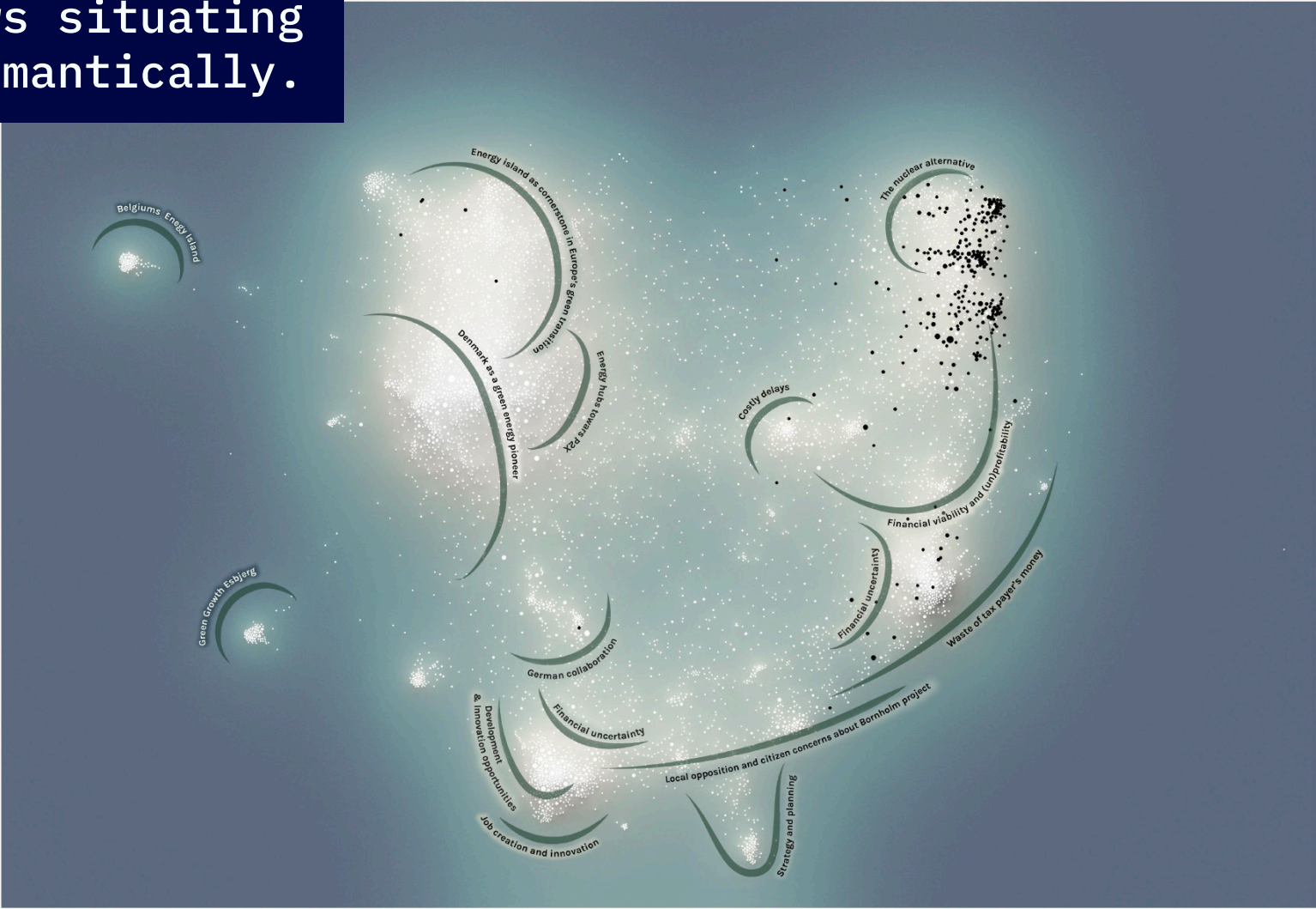
Takeaway:

Do not denounce actors who you think lack objectivity.

Instead, describe objectively the extent of their subjectivity.

How the map supports exploration

The map allows situating statements semantically.



● Issue 9

Is nuclear power an alternative to energy islands?

Several actors argue that nuclear power could be a more cost-effective alternative because it offers a stable energy supply regardless of weather conditions. While nuclear is often criticized for its lengthy construction times and associated costs, energy island projects have faced similar criticism. You could therefore argue that the Danish ban on nuclear power should be lifted to explore it as a future alternative. From that proposition ensues a range of claims and counterclaims about the safety and location of nuclear power plants, including technological advancements to improve safety and handling of nuclear waste.

It helps you profiling specific discourses.

How big is this issue?

Number of statements: **276**
Proportional share of all statements: **4.6%**



Proportional share of all actor statements per year



Top 5 actors addressing issue 9

Ranked by the number of statements per actor on this issue.

*Social Media actors have been excluded from top 5

Mona Juul



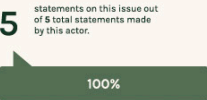
This represents 2.9% of all the statements any actor has made about this issue.

Steffen W. Frølund



This represents 2.2% of all the statements any actor has made about this issue.

Kärnfull Energi

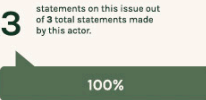


This represents 1.8% of all the statements any actor has made about this issue.



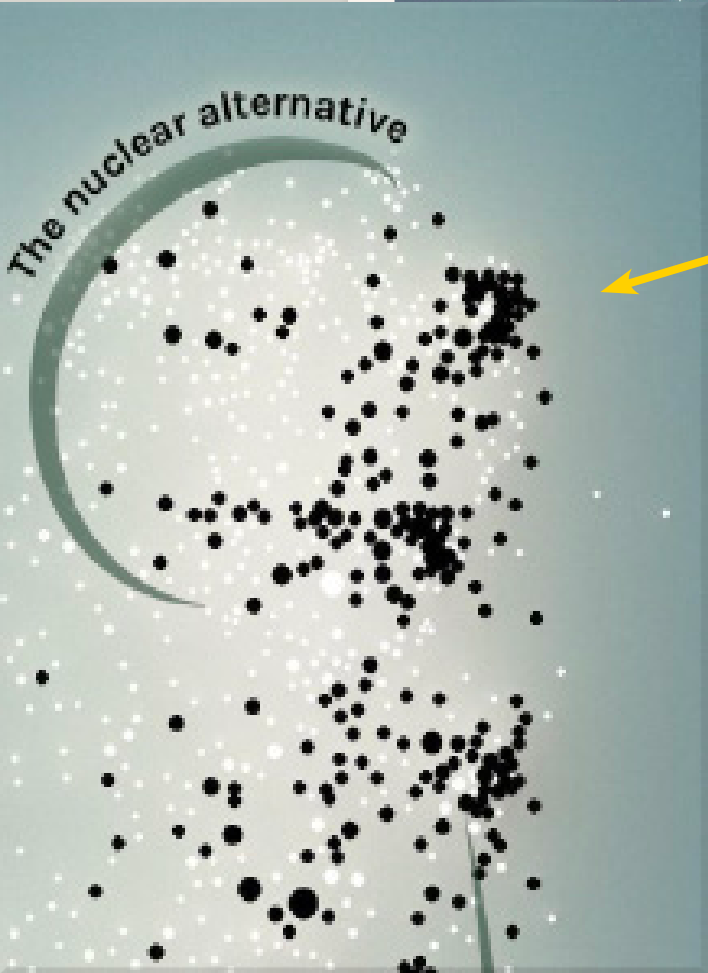
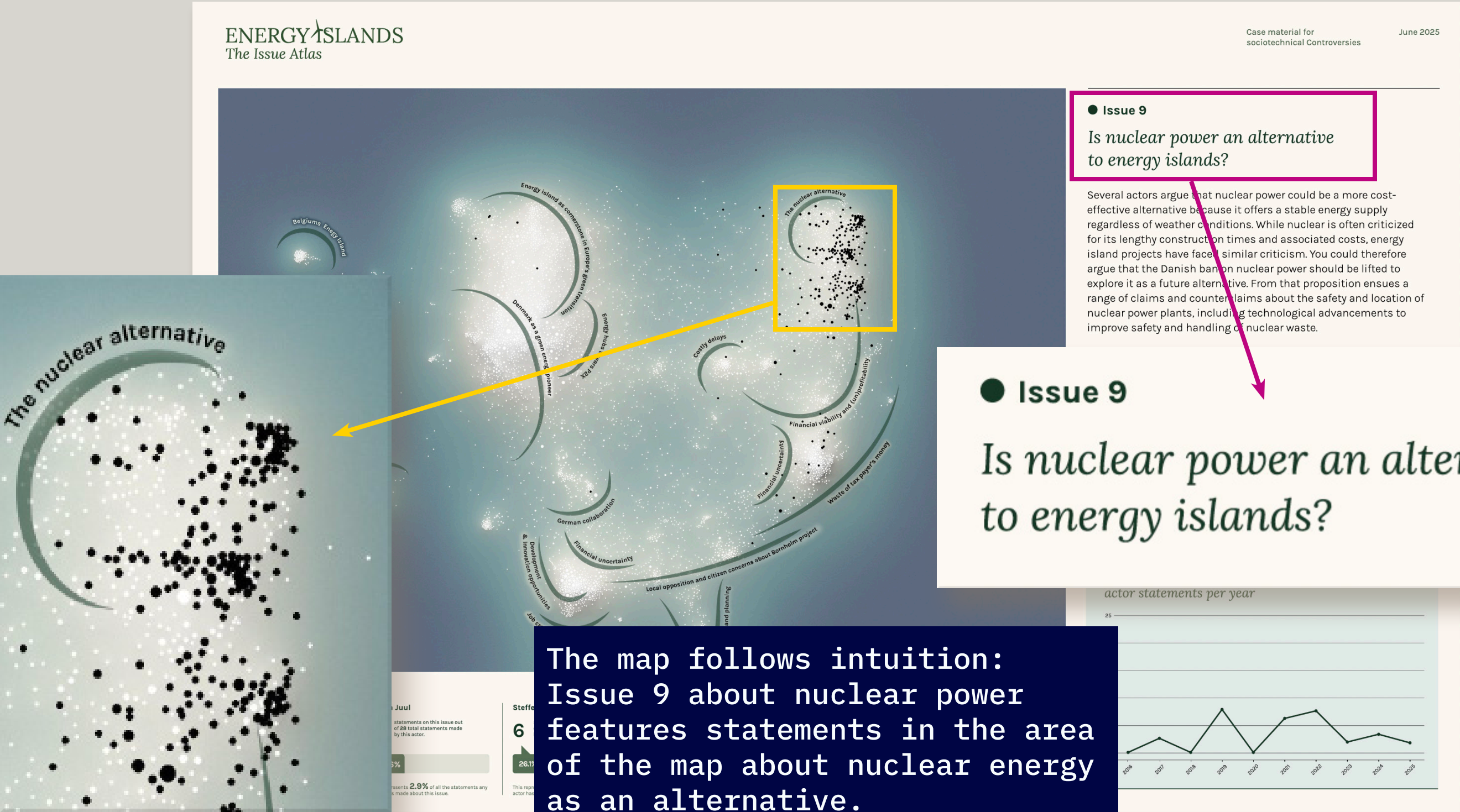
This represents 1.1% of all the statements any actor has made about this issue.

TerraPower



This represents 1.1% of all the statements any actor has made about this issue.

How the map supports exploration



The map follows intuition:
Issue 9 about nuclear power
features statements in the area
of the map about nuclear energy
as an alternative.



Issues: top-down coding

Methodology:

1. Manually identify 16 issues (i.e., problematics).
2. Describe each issue qualitatively.
3. Use AI (large language model) and the descriptions to code each actor statement.

Properties:

- **Non-mutually exclusive:** each statement can be associated with zero, one, or multiple issues.
- **Defined by research:** their existence reflects the interests of researchers (“top-down”).
- **Semantically heterogeneous:** most issues capture a variety of different statements.

Clusters: bottom-up coding

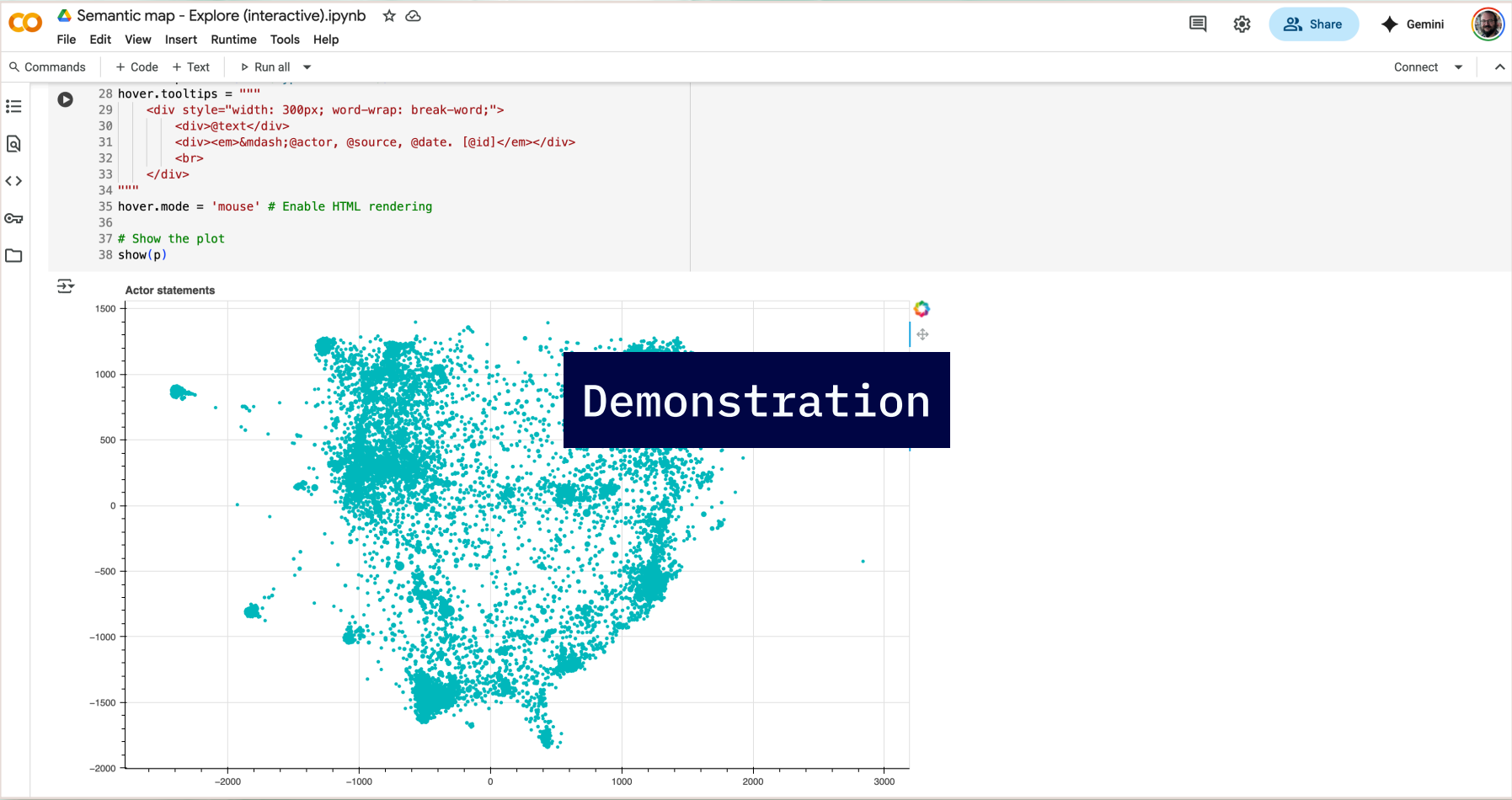
Methodology:

1. Embed the corpus of statements.
2. Identify dense clusters in the vector space.
3. Manually label each cluster.

Properties:

- **Mutually exclusive:** each statement is assigned exactly zero or one cluster, which is often convenient.
- **Defined by actors:** their existence reflects the discourse of actors (“bottom-up”).
- **Semantically homogeneous:** by design, clusters contain similar statements.

How to use the notebooks



THANK YOU FOR YOUR ATTENTION

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